25-Company Validation Run — Plan (PREPARED, NOT LAUNCHED)

Status: this is a proposal. Nothing is launched. No code, replica, or configuration change is made by this document. The run is described so cost/time boundaries and stop rules are agreed *before* any launch. Launch requires an explicit, separate go-ahead.

1. Purpose

Validate the already-accepted worker-isolation build at ~2.5× the ten-company acceptance scale, on the **unchanged** build and the **actual** two-worker configuration, to confirm isolation reliability and pipeline completion hold on a larger, independent roster.

2. Fixed configuration (unchanged from batch 1275)

Parameter	Value
Build / SHA	`b676c2ddf37512dad0e18c6e812a4dc7e58a12a2` (no change)
`WORKER_ISOLATION_ENABLED`	`true` (remains enabled)
Worker replicas	2 (unchanged)
Concurrency / replica	4 → **8 execution slots** total (unchanged)
LLM permit pool	`max = 8` (unchanged)
Framework	18 (“AI Management”, 36 measures)
Workspace	3
Per-job watchdog	existing `JOB_TIMEOUT_MS` / pipeline deadline (unchanged)

No migration, backfill, dedup, `db:push`, tuning, or replica change is part of this plan.

3. Roster

Proposed roster: `list_id = 7` (“25 companies”), exactly 25 members, company_ids 9–30 and 2566–2568 (CSX, Magna, Southern Co., S&P Global, Martin Marietta, Darden, BCE, Stryker, Amphenol, Aflac, Celestica, Dominion, Power Assets, BNP Paribas, ABB, AXA, Otsuka, Aristocrat, Associated British Foods, BT Group, Idemitsu Kosan, Barratt Redrow, Coterra Energy, J M Smucker, FactSet).

This cohort is **independent of the ten-company acceptance set** (147, 148, 158, 257, 1063, 1486, 1533, 2004, 2682, 2696) — no overlap — which strengthens validation by exercising fresh corpora rather than re-scoring known-good companies. It also avoids corpus-pinning to the acceptance set.

Roster must be re-confirmed at launch time. If the list membership has changed, or a different 25-company set is preferred, the roster is finalised then; this plan fixes the *size* (25) and the *source list* as the default, not a frozen id-list.

4. Expected work

- Jobs: **25**, each the full framework-18 pipeline (36 measures).
- Expected measure results: 25 × 36 = **900** scored rows at completion.
- Expected snapshot: one immutable `analysis_results` row, `companies_count = 25`.

5. Cost boundary

Derived from batch 1275 actuals (mean $0.405/company, observed range $0.3651–$0.4520):

Basis	25-company estimate
Central (mean $0.405 × 25)	**≈ $10.1**
Band (min–max per-company × 25)	**$9.1 – $11.3**
Proposed hard cost ceiling	**$14.0** (≈1.4× central)

The $14 ceiling absorbs a heavier/longer roster and more provider fallbacks than the acceptance set saw. **Stop rule S3** (below) enforces it.

6. Time boundary

Ten-company batch 1275 ran 24m 32s wall across effectively 2 slot-waves (8 immediate + 2 queued). 25 jobs over 8 slots ≈ **3.1 waves**. Observed per-job execution ranged 9m 34s – 23m 40s (mean ≈ 15m).

Basis	25-company estimate
~3–4 waves × ~15m mean	**≈ 45 – 70 min** wall
Proposed hard wall-clock ceiling	**90 min**

Per-job watchdog remains the existing deadline (unchanged); the wall-clock ceiling is a *monitoring* stop rule (S2), not a new timeout setting.

7. Stop rules (decided before launch)

Monitoring is read-only observation of the same DB tables and worker logs used for batch 1275. If any rule trips, **stop/cancel the batch, then verify child + permit settlement** (no `[WORKER-ISOLATION]` quarantine residue; `[DIAG][llm-permits]` returns to `held = 0, waiting = 0`) before any further action. Isolation reliability is judged separately from completion throughout.

ID	Condition	Action
S1 — Isolation breach	Any `[WORKER-ISOLATION]` quarantine, withheld-permit, or forced-kill (SIGTERM/SIGKILL) line, **or** permits fail to return to 0 (leak)	**Stop immediately.** Capture logs. Do not retry. This is the primary gate — isolation reliability failure overrides completion.
S2 — Wall-clock overrun	No batch completion by **90 min** wall, or no job state progress for **> 40 min** (stall)	Stop/cancel; verify settlement; report as INCONCLUSIVE (a genuine stop condition is evidence, not a failure).
S3 — Cost overrun	`llm_usage_events` cumulative cost for the batch exceeds **$14.0**	Stop/cancel; verify settlement; report spend at stop.
S4 — Failure rate	**≥ 2** jobs reach `failed`, or any job exhausts retries	Stop/cancel; capture failing attempt_ids + logs; verify settlement.
S5 — Completeness shortfall at completion	Batch reports completed but `measure_scores` < 900 rows, or any company ≠ 36, or any null/abstained not explainable as legitimate "no information"	Do not accept; flag the shortfall; keep snapshot for inspection.
S6 — Provenance/snapshot defect	Snapshot not written, not immutable, or `companies_count ≠ 25`; or attempt_id inconsistency within a company	Flag; do not accept.

Reporting discipline on stop: an S2 timeout or any genuine stop condition is reported as INCONCLUSIVE with the evidence captured, **not** silently as a failure, and completion-time observations (e.g. the parent-RSS spike from §3.3 of the addendum) are recorded as observations, not diagnoses.

8. What to watch (carried from batch 1275 observations)

- **Parent-RSS completion spike** (addendum §3.3): instrument whether it recurs and whether magnitude scales with roster size. Observation only.
- **Browser-render `Cannot fork` pressure** (addendum §3.4): browser-subprocess limits may rise with more concurrent renders at 25 companies. Watch; not a job-level failure unless it blocks completion.
- **Parent RSS vs container memory**: still only parent RSS is instrumented; container/child memory remains uninstrumented (labelling caveat stands).

- **8-slot ceiling**: 25 jobs will queue ~17 behind the first 8; queue-wait tail will be longer than batch 1275. Expected and benign — distinguish queue wait from execution time when reading results.

9. Launch checklist (for the future go-ahead — do NOT act now)

1. Confirm HEAD still `b676c2d…`; confirm 2 replicas × concurrency 4; confirm `WORKER_ISOLATION_ENABLED=true`.
2. Confirm/finalise the 25-company roster (default `list_id = 7`).
3. Confirm framework 18, workspace 3.
4. Start monitoring (read-only) for S1–S6 before enqueue.
5. Enqueue the single 25-company batch. **(This step only on explicit go-ahead.)**
6. On completion or any stop rule: run the same completion-addendum queries (completeness, provenance, snapshot, costs, timing) and the isolation-reliability checks, reported separately.

This plan launches nothing. It fixes scale (25), roster source, cost ceiling ($14), wall ceiling (90 min), and stop rules S1–S6 so a later run is bounded and decision-driven.